29/01/24 A/S

USN | | | | | | | | | | |

[23CSBS101]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

FUNDAMENTALS OF COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Answer All Questions

		PART - A	**05 X 02=10**		
1.	**a.**	Define sequential logic structure.	L1	CO1	**02**
	b.	Define an Array.	L1	CO2	**02**
	c.	What do you mean by String?	L1	CO3	**02**
	d.	What are pointers? State the advantages of pointers.	L1	CO4	**02**
	e.	Write the syntax for the structure to create students details	L1	CO5	**02**
		PART - B	**06 X 05=30**		
2.	**a.**	Explain following for finding the area of the triangle. **i.** Algorithm **ii.** Flowchart **iii.** Pseuocode	L2	CO1	**05**
		OR			
	b.	Describe in brief the features of C programming.	L2	CO1	**05**
3.	**a.**	List out the types of operators in C. Explain one of the types of operators with an example.	L1	CO2	**05**
		OR			
	b.	Write a C program to find the factorial of a given number?	L3	CO2	**05**
4.	**a.**	Explain the following parameter passing methods with an example C program? **i.** Call by value **ii.** Call by reference	L2	CO3	**05**
		OR			
	b.	Describe recursion and Write a C program to compute the sum of n natural numbers using recursion.	L2	CO3	**05**
5.	**a.**	What are the pointers? Explain pointer to pointer with an example C program.	L2	CO4	**05**
		OR			
	b.	Write a C program to add two numbers using pointers.	L3	CO4	**05**
6.	**a.**	What is the Self-referential structure? Explain with an example C program	L2	CO5	**05**
		OR			
	b.	List out the difference between Structure and Union with an example C program	L2	CO5	**05**

7.	a.	Explain the byte ordering methods in detail.	L2	CO1	**05**
		OR			
	b.	Explain about the Standard I/O and formatted I/O functions in C.	L2	CO1	**05**
		PART - C	**06 X 10=60**		
8.	a.	What are the sections of a C Program? Explain with an example C program.	L2	CO1	**10**
		OR			
	b.	State the purpose of using data types in C. Explain the types of data types in C with the neat illustrations.	L2	CO1	**10**
9.	a.	What are the unconditional control structures? Explain with an example C program.	L2	CO2	**10**
		OR			
	b.	Describe looping statements in C. Explain with an example C program.	L2	CO2	**10**
10.	a.	Explain in detail about the storage classes in C with an example.	L2	CO3	**10**
		OR			
	b.	Write a C program to sorting of array elements using pointers.	L3	CO3	**10**
11.	a.	Implement a C program to find the addition of two matrix.	L3	CO4	**10**
		OR			
	b.	Explain subtraction operation on two pointers with an example C Program.	L2	CO4	**10**
12.	a.	Define an array of structure and explain it with an example	L2	CO5	**10**
		OR			
	b.	What are the file handling operations in C. Explain with an example C program.	L2	CO5	**10**
13.	a.	Explain any of the two user defined functions in C to check if a number is even or odd.	L2	CO3	**10**
		OR			
	b.	Explain about the scope rule and the types of variables used in C with the neat illustrations.	L2	CO2	**10**